Please read the instructions below before the exam starts, and listen carefully to the instructions given by the invigilators.

You may read the question paper before the exam starts but you must not write anything until the invigilator says that you may start.

You will be given five minutes after the exam has finished to complete the front of any answer books used.

May/June 2024

EC116 2023/4 A 800
EC116 2022/3 A 800

Calculators Permitted

UNIVERSITY OF READING

INTRODUCTORY MATHEMATICS FOR ECONOMICS 1 (EC116)

Two hours

Answer **two** questions only in Section A (60 marks);

Answer **All** questions in Section B (40 marks)

SECTION A

1. Consider the following model of the economy:

$C = 200 + 0.6 (Y – T)$
$I = 200$
$G = 200$
$T = 0.1 Y + 50$

(a) Compute the equilibrium level of income of the economy and show your answer graphically.

(8 marks)

(b) At the equilibrium level of income, what is the value of the government budget surplus?

(7 marks)

(c) Sketch the consumption function, *C*, by using the slope-intercept approach.

(7 marks)

(d) Suppose *I* decreases by 100, calculate its effect on equilibrium income.

(8 marks)

2. A monopolist's demand and total cost functions of a good are $P = -Q + 2$ and $C = Q + Q^2$, respectively.

   (a) Derive the total revenue (R) and profit (π) functions.

   (10 marks)

   (b) Sketch the cost function (for $Q \geq 0$ only; put C and Q on vertical and horizontal axis, respectively).

   (10 marks)

   (c) Calculate the profit maximising output Q* and the maximum profit π* using differentiation. (Hint: verify the result with the second-order derivative.)

   (10 marks)

3. Given the following demand and supply functions $P = 500 - \mathrm{Q}$ and $\mathrm{P} = 100 + 2Q$, respectively,

   (a) Calculate the equilibrium price and quantity in the market. Sketch the demand curve, supply curve and your answers to part (a) in one diagram (for $P \geq 0$ and $Q \geq 0$ only; put P and Q on vertical and horizontal axes, respectively).

   (12 marks)

   (b) Calculate the price elasticities of demand and supply at the market equilibrium using differentiation.

   (8 marks)

   (c) Suppose the government imposes a per unit tax (t) on the good and t is 20. Show the effect of the tax on P and Q numerically and graphically (Hint: you do not need to calculate the price elasticities).

   (10 marks)

SECTION B

4. (a) Differentiate the following equations and find their slopes at x=1.

(i) $y = (2x + 3x^2)^2$

(ii) $y = 3x - ln(x^3)$

(iii) $y = \frac{2x^2}{e^x}$

(12 marks)

(b) Find the stationary points of the following equation. Use the second-order derivative method to check whether they are local maximum or minimum.

$$y = -3x^3 + 2x^2 + x$$

(18 marks)

(c) Use the rules of logs to expand the following:

$$\ln \sqrt{\frac{xz^4}{y^6}}$$

(5 marks)

(b) Use the rules of logs to express the following as a single logarithm:

5 ln x - 3 ln z + ln y

(5 marks)

(End of Question Paper)